

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2022

TEST DATE: 17-04-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

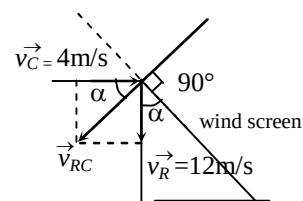
SECTION – A

1. B

Sol. Velocity of rain with respect to car $\vec{v}_{RC} = \vec{v}_R - \vec{v}_C$ should be perpendicular to the wind screen.
From figure,

$$\tan \alpha = \frac{v_r}{v_c} = \frac{12}{4}$$

$$\alpha = \tan^{-1}(3)$$



2. A

Sol. $PV = RT$ for 1 mole

$$W = \int PdV = \int \frac{RT}{V} dV$$

$$V = CT^{2/3}$$

$$\therefore dV = \frac{2}{3} CT^{-1/3} dT \quad \text{or} \quad \frac{dV}{V} = \frac{2}{3} \frac{dT}{T}$$

$$\therefore W = \int_{T_1}^{T_2} RT \left(\frac{2}{3} \right) \frac{dT}{T} = \frac{2}{3} R(T_2 - T_1) = 166.2 \text{ J}$$

3. B

Sol. $v_b = 20 \text{ m/s} < \sqrt{5g\ell}$

$\therefore$ Particle leaves the circular path.

$$20 = \sqrt{g\ell(2 + 3\cos\phi)} \Rightarrow \cos\phi = \frac{2}{3}$$

$$\text{Velocity at top} = \left(\sqrt{g\ell\cos\phi}\right)\cos\phi = \frac{20}{3}\sqrt{\frac{2}{3}} \text{ m/s}$$

4. A

Sol. $mR\omega^2 \propto \frac{1}{R^{1/n}}$

5. C

Sol. Volume of ball $V = \frac{m}{\rho}$

Acceleration of ball inside the liquid

$$a = \frac{F_{\text{net}}}{m} = \frac{\text{upthrust} - \text{weight}}{m}$$

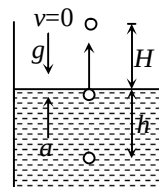
$$\text{or } a = \frac{\left(\frac{m}{\rho}\right)(5\rho)(g) - mg}{m} = 4g \text{ (upwards)}$$

$\therefore$ Velocity of ball while reaching at surface

$$v = \sqrt{2ah} = \sqrt{8gh}$$

$\therefore$ The ball will jump to a height

$$H = \frac{v^2}{2g} = \frac{8gh}{2g} = 4h$$



6. C

Sol. At the top

$$T + mg = \frac{mv^2}{L}$$

$$T < 10mg$$

$$v < \sqrt{11gL}$$

7. A

Sol. $F = -mg \sin \theta$

$$F = -mg \sin(\alpha + \beta)$$

$$ma = -mg(\alpha + \beta)$$

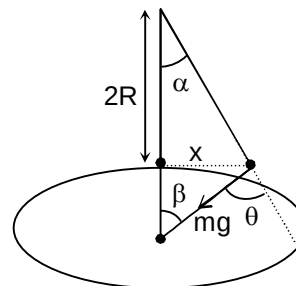
$$a = -g \left[\frac{x}{2R} + \frac{x}{R} \right]$$

$$a = -\left(\frac{3g}{2R}\right)x \quad \dots(i)$$

$$a = -\omega^2 x \quad \dots(ii)$$

from equations (i) and (ii)

$$T = 2\pi\sqrt{\frac{2R}{3g}}$$



8. D

Sol. For fifth overtone $\frac{5\lambda}{2} + \frac{\lambda}{4} = 33 \text{ cm}$, $\lambda = 12 \text{ cm}$
at a distance 18 cm ($= 3\lambda/2$) from closed end node is formed

9. B

Sol. Here $u = v_0$, $\omega_0 = -\frac{v_0}{2R}$

At pure rolling,

$$v = v_0 - \left(\frac{F_1}{m}\right)t \text{ and } \frac{v}{R} = -\frac{V_0}{2R} + \left(\frac{F_1}{mR}\right)t \text{ (in pure rolling } v = R\omega) \left(\alpha = \frac{\tau}{I} = \frac{F_1 R}{mR^2}\right)$$

$$\Rightarrow v_0 - v = v + \frac{v_0}{2}$$

$$\Rightarrow 2v = \frac{v_0}{2} \Rightarrow v = \frac{v_0}{4}$$

10. D

Sol. By using $E = \frac{V}{d} = \frac{0.6}{10^{-6}} = 6 \times 10^5 \text{ V/m}$

11. D

Sol. $P = \sigma AT^4$

$$\Rightarrow \frac{P_A}{P_B} = \frac{A_A}{A_B} \left(\frac{T_A}{T_B}\right)^4 \Rightarrow T_B = T_A (2)^{3/4}$$

$$\text{as } \lambda_m T = \text{constant} \Rightarrow \frac{\lambda_A}{\lambda_B} = \frac{T_B}{T_A}$$

$$\Rightarrow \lambda_B = 5000(2)^{\frac{3}{4}} \text{ \AA}$$

12. B

Sol. $\frac{1}{\lambda_{k_\alpha}} = RZ^2 \left(\frac{1}{1^2} - \frac{1}{2^2}\right)$ and $\frac{1}{\lambda_{k_\beta}} = RZ^2 \left(\frac{1}{1^2} - \frac{1}{3^2}\right)$

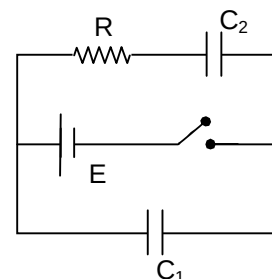
Dividing we get,

$$\lambda_{k_\beta} = 0.27 \text{ \AA}$$

13. B

Sol. Rearranging the circuit, we observed that C_1 is joined directly to the cell and acquires its full charge when S is closed. It plays no part in the charging of C_2 through R.

$$\text{So, } q_2 = Q_0(1 - e^{-t/\tau})$$



14. C

Sol. The current lags the emf by $\pi/2$, so the circuit should contain only an inductor.

15. D

Sol. $x = A \sin \frac{2\pi}{T} t$; for $x = \frac{A}{2}$

$$\Rightarrow \frac{A}{2} = A \sin \frac{2\pi}{T} t$$

Solving, $t = \frac{T}{12}$

16. B

Sol. $I_1 = I$ and $I_2 = 4I$

$$\therefore I_{\text{net}} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$\therefore I_A = I + 4I = 5I \quad (\because \phi_A = \pi/2)$$

and $I_B = I + 4I - 2\sqrt{4I \cdot I} = I \quad (\phi_B = \pi)$

$$\Rightarrow I_A - I_B = 4I$$

17. A

Sol. $I = I_{\text{max}}(1 - e^{-t/\tau})$

18. C

Sol. $m_{\text{ice}} S_{\text{ice}}(10) + m_{\text{ice}} L_{\text{ice}} = ML_V + MS_{\text{water}}(100)$
 $\Rightarrow 3200(0.5)(10) + 3200(80) = m[540 + 100]$
 $\Rightarrow m = 425 \text{ gm}$

19. D

 Sol. For maximum current, net resistance of cell must be equal to 2.5Ω

i.e. $\frac{n(0.5)}{m} = 2.5 \quad \dots(i)$

and $m \times n = 45$

solving, we get $n = 15$, $m = 3$

20. C

Sol. $F = \frac{kq^2}{r^2}; \left(k = \frac{1}{4\pi\epsilon_0} \right)$

$$\Rightarrow \frac{kq^2}{r^2} = \frac{mv^2}{R_c}$$

$$R_c = \frac{mv^2 r^2}{kq^2}; R_c = \frac{4\pi\epsilon_0 V^2 r^2 m}{q^2}$$

SECTION – B

21. 00002.50

Sol. Taking moments about the edge about which the cube is likely to tilt,

$$F \times \frac{4}{5}a = mg \times \frac{a}{2}$$

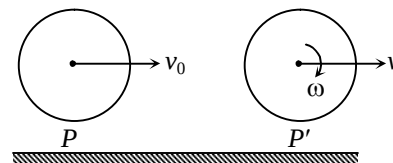
$$\therefore F = \frac{5}{8}mg$$

22. 00000.50

Sol. Let v be the velocity of centre of mass of ring just after the impulse is applied and v is its velocity in pure rolling.

$$J = mv_0$$

$$v_0 = \frac{J}{m}$$



Conserving angular momentum about point of contact with ground

$$mv_0 r = mvr + I_{cm} \omega$$

$$\Rightarrow v = \frac{v_0}{2} = \frac{J}{2m}$$

23. 00004.00

Sol. $\frac{1}{2}mv_{\max}^2 = 2J$, $v_{\max} = 2 \text{ m/s} = A\omega$, $v_{\text{avg}} = \frac{2A}{T/2} = \frac{4A}{T} = \frac{4A\omega}{2\pi} = \frac{4}{\pi} \text{ m/s}$

24. 00004.33

Sol. $m_1 = 1 \text{ kg}$, $m_2 = 2 \text{ kg}$, $u_1 = 3 \text{ ms}^{-1}$, $u_2 = 2 \text{ ms}^{-1}$

$$\text{Initial momentum} = \sqrt{(m_1 u_1)^2 + (m_2 u_2)^2} = \sqrt{9 + 16} = 5 \text{ kg ms}^{-1}$$

$$\text{If combined velocity is } v, (m_1 + m_2)v = 5, \quad v = \frac{5}{3} \text{ ms}^{-1}$$

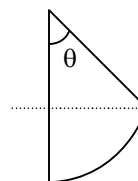
$$\text{Heat liberated} = \text{loss in kinetic energy} = \frac{1}{2}m_1 u_1^2 + \frac{1}{2}m_2 u_2^2 - \frac{1}{2}(m_1 + m_2)v^2 = \frac{13}{3} J$$

25. 00000.25

Sol. $\sin \theta = \frac{\frac{mv}{\sqrt{2}qB}}{\frac{mv}{qB}} = \frac{1}{\sqrt{2}}$

$$\Rightarrow \theta = 45^\circ$$

$$t = \frac{T}{8} = \frac{\pi m}{4qB}$$



26. 00100.00

Sol. At maximum depth the ray graze the surface (i.e. the angle made by the ray with normal will become 90°)

Applying Snell's law

$$1 \times \sin 45^\circ = \left(\sqrt{2} - \frac{1}{\sqrt{2}} x \right) \sin 90^\circ$$

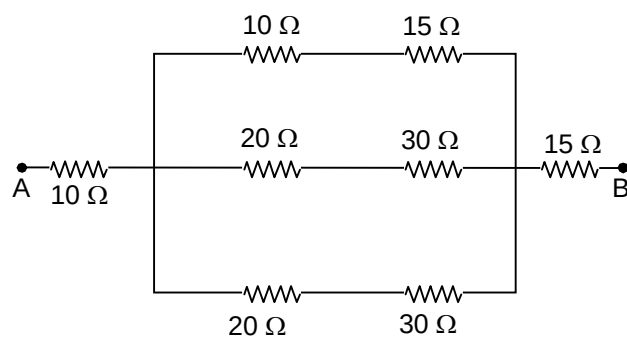
$$\Rightarrow \sqrt{2} - \frac{1}{\sqrt{2}} x = \frac{1}{\sqrt{2}} \text{ or } x = 1 \text{ m}$$

27. 00002.20

Sol. Energy released = $(80 \times 7 + 120 \times 8 - 200 \times 6.5) = 220 \text{ MeV}$

28. 00037.50

Sol. equivalent circuit is
 $= 37.5 \Omega$



29. 00012.00

Sol. $(\chi) = \frac{\text{Intensity of magnetisation}(I)}{\text{Magnetizing field}(H)}$

Or, $I = \chi H$

30. 00004.00

Sol. $eV_0 = K. \Rightarrow V_0 = \frac{K.E_{\max}}{e}$

Chemistry**PART – B****SECTION – A**

31. B

Sol. If n – quantum number, then, $n\lambda = 2\pi r$ $[\lambda = \text{de-Broglie's wavelength}]$

$$r = \frac{n^2 a_0}{Z}$$

$$n\lambda = 2\pi \frac{n^2 a_0}{Z}$$

$$\lambda = \frac{2\pi n a_0}{Z} \quad Z = 3 \text{ for } \text{Li}^{2+}$$

$$\lambda = 2\pi a_0$$

$$2\pi a_0 = \frac{2\pi n a_0}{3}$$

$$\Rightarrow n = 3$$

32. C

Sol. 500 ml of water contains 24 mg of MgSO_4 $\therefore$ 1000 ml of water contains $24 \times 2 = 48$ mg of MgSO_4

$$48 \text{ mg } \text{MgSO}_4 = \frac{48}{1000} \text{ gm } \text{MgSO}_4$$

$$= 0.048 \text{ gm } \text{MgSO}_4$$

$$\text{Number of moles of } \text{MgSO}_4 = \frac{0.048}{120}$$

$$= 4 \times 10^{-4} \text{ mole}$$

i.e. number of moles of CaCO_3 (hypothetical) would be 4×10^{-4} in 1 litre.

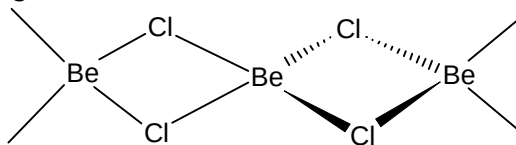
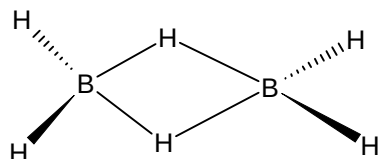
$$\text{Mass of } \text{CaCO}_3 = 4 \times 10^{-4} \times 100 = 0.04$$

$$= 40 \text{ mg.}$$

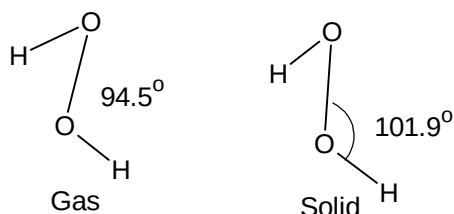
 $\therefore$ Hardness in ppm = 40

33. C

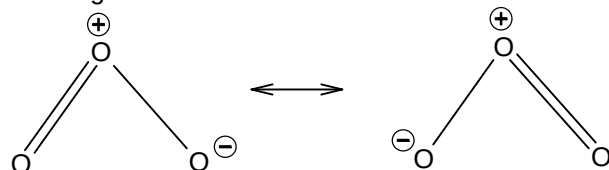
Sol.

Be is sp^3 hybridised.

The bridge H forms longer bond



The angle between H – O and O – O are not the same in solid and gas.



O_3 is polar, as there is a charge separation and the molecule is bent.

34. D

Sol. ${}^2_1\text{H}$ and ${}^3_2\text{He}$ are isotones as they have same number of neutrons but different mass number.

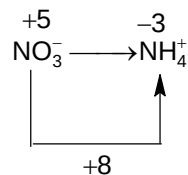
${}^{37}_{17}\text{Cl}$ and ${}^{39}_{18}\text{Ar}$ are isodiaphers as they have same excess of neutron over protons ($A - 2Z$).

${}^7_3\text{Li}$ and ${}^7_4\text{Be}$ are isobars as they have same mass number but different atomic numbers. Mirror nuclei have equal mass numbers but their atomic numbers differ by 1.

${}^2_1\text{H}$ and ${}^3_2\text{He}$ can not be mirror nuclei as mass numbers are different.

35. D

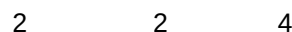
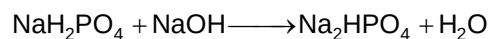
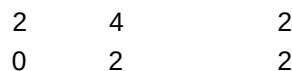
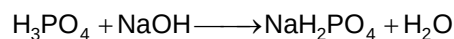
Sol.



36. A

Sol. 10 millimoles of $\text{H}_3\text{PO}_4 = 20 \times \frac{1}{10} = 2$

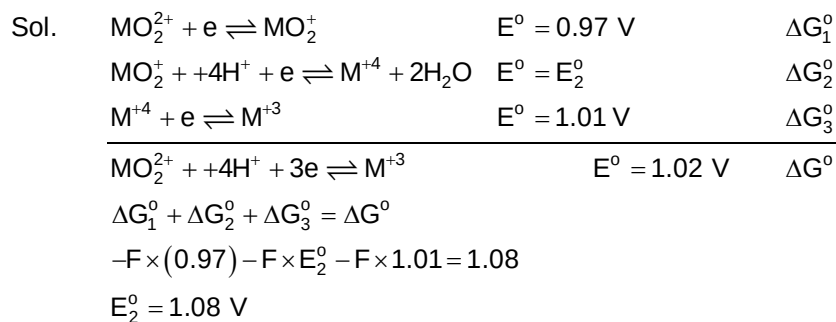
Milimoles of $\text{NaOH} = 4$



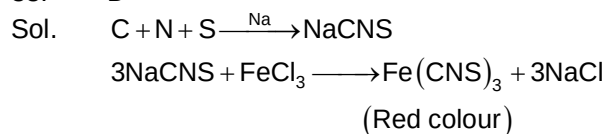
i.e. the resulting solution contains 4 milimoles of Na_2HPO_4 .

$$\begin{aligned}
 \text{i.e. pH of the solution} &= \frac{\text{pK}_2 + \text{pK}_3}{2} \\
 &= \frac{7 + 13}{2} = 10
 \end{aligned}$$

37. B



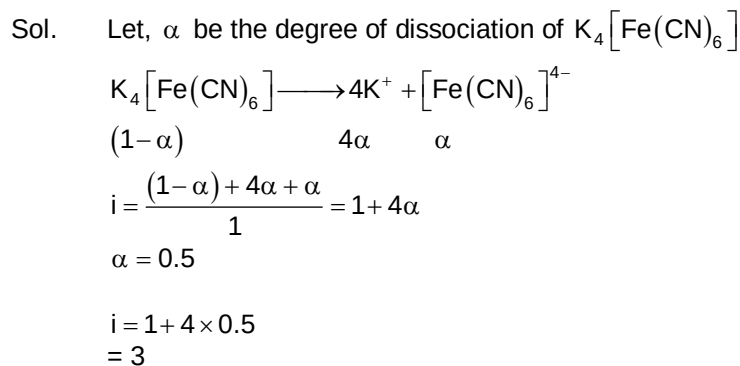
38. D



39. D

Sol. $n = 3 \quad P_1 = 10 \text{ atm} \quad T_1 = 300 \text{ K}, \quad C_p = 29.13 \text{ JK}^{-1}$
 $W_{\text{irr}} = -P_2(V_2 - V_1) = nC_v(T_2 - T_1)$
 $-P_2 \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right) = nC_v(T_2 - T_1)$
 $-R \left(T_2 - T_1 \frac{P_2}{P_1} \right) = (C_p - R)(T_2 - T_1)$
 $-8.314 \left(T_2 - 300 \times \frac{1}{10} \right) = (29.13 - 8.314)(T_2 - 300)$
 $T_2 = 222.94 \text{ K} \approx 223 \text{ K}$

40. C



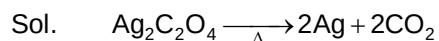
41. A

Sol. Factual

42. B

Sol. CN^- is a strong field ligand. Greater be the oxidation state, greater be Δ_o .

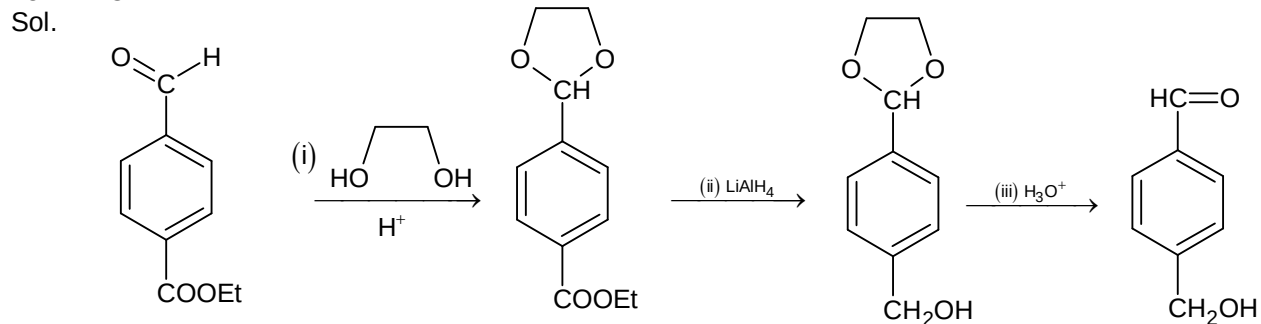
43. B



44. D

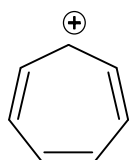
Sol. Planarity at the bridgehead position is not permitted. However, the flexibility of the molecule is increased as number of carbon atoms in the bridge increases.

45. C

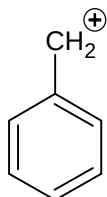


46. D

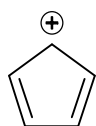
Sol. Greater be the stability of carbocation, greater be the rate of solvolysis.



Aromatic

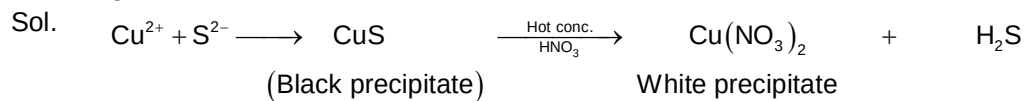


Stabilised through +R effect of benzene ring.

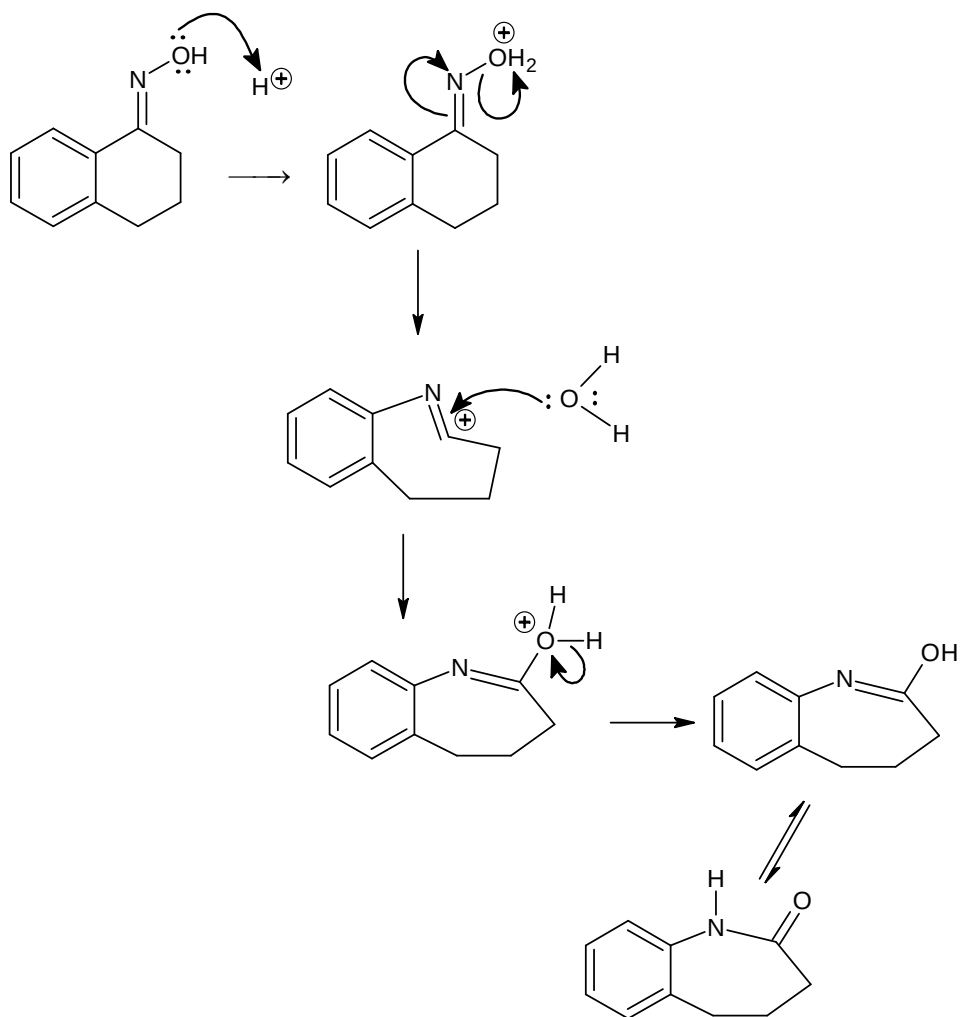


Anti-aromatic.

47. C

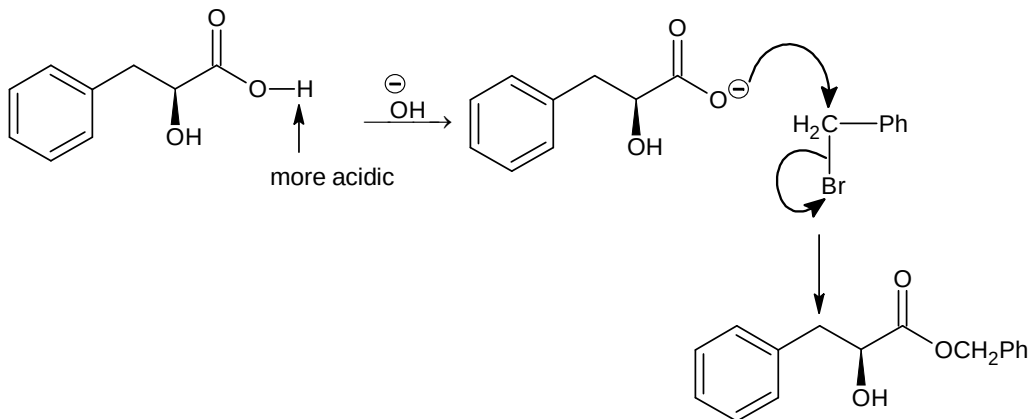


48. D
Sol.



49. A
Sol. O is more electronegative than N. In R, the +R effect of $-\text{OCH}_3$ reduces reactivity of carbonyl group.

50. A
Sol.



SECTION – B

51. 00003.22

Sol. $(PH)_I = \frac{2.19 + 4.25}{2}$
 $= 3.22$

52. 00004.98

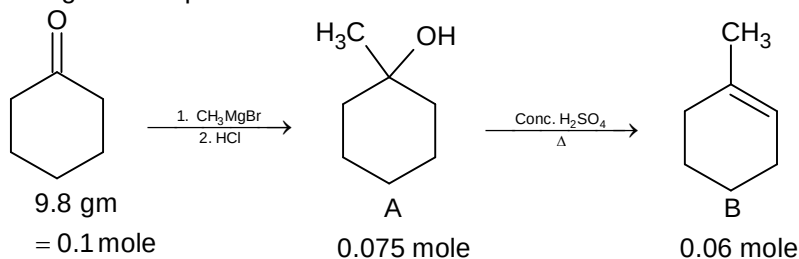
Sol. $n = \frac{8.8}{44} = 0.2$ $T = 273 + 31.1 = 304.1$
 $P = 1 \text{ bar} \approx 1 \text{ atm}$
 $V = \frac{nRT}{P} = \frac{0.2 \times 0.082 \times 304.1}{1}$
 $= 4.98 \text{ l}$

53. 00016.083

Sol. $P_{\text{total}} = 0.2 \text{ bar}$
 At equilibrium, $P_{\text{CO}_2} = P_{\text{H}_2\text{O}} = 0.1 \text{ bar}$
 $K_P = P_{\text{CO}_2} \cdot P_{\text{H}_2\text{O}} = (0.1)^2 = 0.01 \text{ bar}$
 $\Delta G^\circ = -RT \ln K_{\text{eq}} = -8.314 \times 420 \times 2.303 \log(0.01)$
 $= 16.083 \text{ kJ.}$

54. 00005.76

Sol. 9.8 gm X was present



Amount of B formed = $0.06 \times 96 = 5.76 \text{ gm}$

55. 00000.39

Sol. $K_a = (HA) = 10^{-7}$, $K_b (\text{BOH}) = 10^{-5}$

$$E_{\text{cell}} = \frac{0.06}{2} \log \frac{[H^+]_{\text{RHS}}^0 (P_{\text{H}_2})_{\text{LHS}}}{[H^+]_{\text{LHS}}^2 (P_{\text{H}_2})_{\text{RHS}}}$$

$$[OH^-] = \sqrt{K_b C} \quad [OH^-] = 10^{-3}$$

$$[H^+]_{\text{LHS}} = 10^{-11}$$

$$[H^+]_{\text{RHS}} = \sqrt{K_a C} = 10^{-4}$$

$$E_{\text{cell}} = \frac{0.06}{2} \log \frac{(10^{-4})^2 \times 0.1}{(10^{-11})^2} = 0.39 \text{ V}$$

56. 00006.46

Sol. $\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$

$$\Rightarrow \frac{n^2 a}{V^2} = \frac{nRT}{V - nb} - P$$

$$\Rightarrow a = \left(\frac{nRT}{V - nb} - P\right) \frac{V^2}{n^2}$$

$$V = 4\ell, P = 11 \text{ atm}, n = 2, b = 0.05, R = 0.082$$

$$a = 6.46$$

57. 00055.87

Sol. $k_1 = \frac{0.7}{20} \quad T_1 = 300 \text{ K}$

$$k_2 = \frac{0.7}{5} \quad T_2 = 320 \text{ K}$$

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$E_a = \frac{RT_1 T_2}{(T_2 - T_1)} \ln \frac{k_2}{k_1}$$

$$= \frac{8.314 \times 320 \times 300}{20} \times \ln 4$$

$$= 55.87 \text{ kJ/mol}$$

58. 00101.785

Sol. For Na_3PO_4 $i = 1 + 3\alpha = 1 + 3 \times 0.5 = 2.5$

$$\text{MgSO}_4 \quad i = 1.6$$

100 gm solution contains 8.2 gm Na_3PO_4 and 12gm MgSO_4

$$\Delta T_b = iK_b m$$

$$= 0.5 \left[\frac{\frac{8.2}{164} \times 2.5 + \frac{12}{120} \times 1.6}{79.8} \right] \times 1000$$

$$= 1.785^\circ\text{C}$$

$$\text{Boiling point} = 100 + 1.785 = 101.785^\circ\text{C}$$

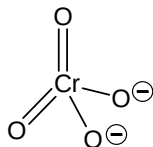
59. 00135.96

Sol. $a^3 = \frac{Z}{d} \left(\frac{M}{N_A} \right) = \frac{2}{10.3} \times \frac{96}{6 \times 10^{23}}$

$$a = 314 \text{ pm}, \quad r = \frac{\sqrt{3}a}{4} = 135.96 \text{ pm}$$

60. 00002.66

Sol.



$$x = 4 \quad \frac{2x}{3} = 2.66$$

Mathematics

PART – C

SECTION – A

61. D

Sol. Let $x - 3 = X$; $y - 1 = Y$, then $|X + Y| = 5$ and $|X| + |Y| = 5$
Hence, total 12 ordered pairs

62. B

Sol. Let $y = \lim_{n \rightarrow \infty} \left(\frac{\ln}{n^n} \right)^{\frac{3n^3+4}{4n^4-1}}$

$$\log y = \lim_{n \rightarrow \infty} \frac{3n^3+4}{4n^4-1} \log \left(\frac{\ln}{n^n} \right) = \frac{3}{4} \int_0^1 \log x \, dx = -\frac{3}{4}$$

$$\therefore y = e^{-\frac{3}{4}}$$

63. C

Sol. $f(x) = 1 + Ax + Be^x$ where $A = \int_0^1 e^y f(y) \, dy$; $B = \int_0^1 y f(y) \, dy$

$$A = \int_0^1 e^y (1 + Ay + Be^y) \, dy ; B = \int_0^1 y (1 + Ay + Be^y) \, dy$$

$$A = -\frac{3}{2} ; B = -\frac{2}{e+1}$$

$$h'(x) > 0 \Leftrightarrow f'(x) + 3 > 0 \Leftrightarrow -\frac{3}{2} - \frac{2e^x}{e+1} + 3 > 0$$

$$e^x < \frac{3}{4}(e+1) ; \frac{4}{3}e^x < (e+1) \Leftrightarrow \left[\frac{4}{3}e^k \right] = 3$$

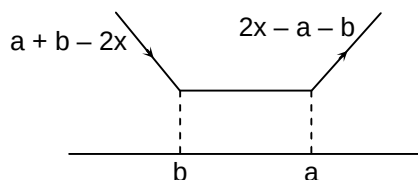
64. B

Sol. $f(x) = |x - a| + |x - b|$
 $f(0) = f(1) = f(-1)$

Let $a > b$

$f(x)$ is constant in $[b, a]$

$b \leq -1$ and $a \geq 1$ minimum $|a - b| = 2$



65. C

Sol. $\begin{vmatrix} p & b & c \\ a-b & q-b & 0 \\ 0 & b-q & r-c \end{vmatrix} = 0$ ($R_3 \rightarrow R_3 - R_2$ and $R_2 \rightarrow R_2 - R_1$)

$$(p-a)(q-b)(r-c) \begin{vmatrix} p & b & c \\ p-a & q-b & r-c \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$

$$(p-a)(q-b)(r-c) \left(\frac{p}{p-a} + \frac{b}{q-b} + \frac{c}{r-c} \right) = 0$$

$$\Rightarrow \frac{p}{p-a} + \frac{b}{q-b} + \frac{c}{r-c} = 0 ; \frac{p}{p-a} + \left(\frac{b}{q-b} + 1 \right) + \left(\frac{c}{r-c} + 1 \right) = 2$$

$$\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = 2$$

66. C

Sol. Following cases ($x \rightarrow$ stands for H or T any one)

$$\underset{\text{m times}}{(\text{HH} \dots \text{H})} \times x \times x \times x \dots = \left(\frac{1}{2} \right)^m \times 1$$

$$\underset{\text{m times}}{\text{T}(\text{HHH} \dots)} \times x \times x \times x \times x \dots = 1 \times \frac{1}{2} \times \left(\frac{1}{2} \right)^m \times 1$$

$$x \text{ T}(\text{HHH} \dots \text{m times}) \times x \times x \dots = 1 \times \frac{1}{2} \times \left(\frac{1}{2} \right)^m \times 1$$

:

$$x \times x \times x \dots \underset{\text{m times}}{\text{T}(\text{HHH} \dots)} = 1 \times \left(\frac{1}{2} \right)^m = \left(\frac{1}{2} \right)^m + n \left(\frac{1}{2} \right)^{m+1} = \left(\frac{1}{2} \right)^m \left[1 + \frac{n}{2} \right] = \frac{n+2}{2^{m+1}}$$

67. B

$$\text{Sol. } (\alpha + \beta + \gamma) \left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \right) = 3 + \frac{\alpha}{\beta} + \frac{\alpha}{\gamma} + \frac{\beta}{\alpha} + \frac{\gamma}{\alpha} + \frac{\beta}{\gamma} + \frac{\gamma}{\beta}$$

$$\Rightarrow (\alpha + \beta + \gamma) \left(\frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} \right) = 3 + \sum \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right)$$

$$0 = 3 + \sum \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) ; \sum \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) = -3$$

68. C

$$\text{Sol. Probability matrix is symmetric} = \frac{7^6}{7^9} = \frac{1}{7^3}$$

$$\text{Probability matrix is skew symmetric} = \frac{7^3}{7^9} = \frac{1}{7^6}$$

One matrix containing all 0 is common in both

$$\text{Required probability} = \frac{1}{7^3} + \frac{1}{7^9} - \frac{1}{7^9}$$

69. C

$$\text{Sol. } f'(x) = g(x)$$

$$g'(x) = f(x^2) = f''(x) \text{ let } y = f(x^3)$$

$$\frac{dy}{dx} = f'(x^3) 3x^2$$

$$\frac{d^2y}{dx^2} = 3[3x^2 \cdot x^2 f''(x^3) + 2xf'(x^3)] = 9x^4 f''(x^3) + 6xf'(x^3) = 9x^4 f(x^6) + 6xg(x^3)$$

$$k + p + a + b + c = 9 + 4 + 6 + 6 + 3$$

70. B

Sol. $(AB)(AB) = A(BA)B = A^3B^2$
 $AB AB AB = A^3 B^2 AB = A^3 B(BA)B = A^3 B A^2 B B = A^3 (BA) AB B = A^3 A^2 (BA) B B = A^7 B^3$
 $(AB)^n = A^{(2^n-1)} B^n$. If $n = 10 \Rightarrow k = (2^{10} - 1) = 1023$

71. A

Sol. Equation of two chords are $y = m(x - ae)$ (1)

$$y = -\frac{1}{m}(x + ae) \quad \text{..... (2)}$$

Multiply equation (1) and (2), we get $x^2 + y^2 = a^2 e^2 \Leftrightarrow x^2 + y^2 = a^2 - b^2$
 Which is director circle of the hyperbola

72. C

Sol. Tangent at (1, 2)

$$2y = 2(x + 1)$$

$$y = x + 1$$

Equation of circle

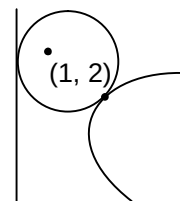
$$(x - 1)^2 + (y - 2)^2 + \lambda(x - y + 1) = 0 \quad \text{..... (1)}$$

Put $x = -1$

$$y^2 - 4y - \lambda y + 8 = 0$$

$$D = 0 \Rightarrow (\lambda + 4)^2 = 32$$

$$y = \frac{(\lambda + 4)}{2} \Rightarrow y = \pm \frac{4\sqrt{2}}{2} = \pm 2\sqrt{2}$$



73. A

Sol. Let x_1, x_2, x_3 are roots of equation A.M. $\frac{x_1 + x_2 + x_3}{3} = 2$; H.M $\frac{3}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3}} = 2$

$$A.M = H.M ; x_1 = x_2 = x_3 = 2$$

$$\text{Hence, } \sin^{-1}\left(\frac{1}{x_1} + \frac{1}{x_2}\right) + \cos^{-1}\left(\frac{1}{x_2} + \frac{1}{x_3}\right) - \tan^{-1}\left(\frac{1}{x_2} + \frac{1}{x_1}\right) = \frac{\pi}{2} + 0 - \frac{\pi}{4} = \frac{\pi}{4}$$

74. C

Sol. $\therefore (2010)^{\frac{1}{2010}} < (2009)^{\frac{1}{2009}} ; (2010)^{2009} < (2009)^{2010}$

$$\therefore \lim_{n \rightarrow \infty} \left[\left((2009)^{2010} \right)^n + \left((2010)^{2009} \right)^n \right]^{\frac{1}{n}} ; \lim_{n \rightarrow \infty} (2009)^{2010} \left[1 + \left(\frac{(2010)^{2009}}{(2009)^{2010}} \right)^n \right]^{\frac{1}{n}}$$

$$(2009)^{2010} = a^b ; a = 2009 ; b = 2010 \therefore a - b = -1$$

75. C

Sol. $g'(y) = \frac{1}{f'(x)} ; g'(8) = \frac{1}{f'(1)} = \left[\frac{1}{3x^2 + 4x + 4 + \left(\cos \frac{\pi x}{2} \right) \frac{\pi}{2}} \right]_{x=1} = \frac{1}{11}$

76. A

Sol. $\cot \alpha_1 \cdot \cot \alpha_2 \cdot \cot \alpha_3 \dots \cot \alpha_n = 1$

$$\cos \alpha_1 \cdot \cos \alpha_2 \cdot \cos \alpha_3 \dots \cos \alpha_n = \sin \alpha_1 \cdot \sin \alpha_2 \dots \sin \alpha_n$$

$$\text{Let } y = \cos \alpha_1 \cdot \cos \alpha_2 \dots \cos \alpha_n$$

$$y = \sin \alpha_1 \cdot \sin \alpha_2 \dots \sin \alpha_n$$

$$y^2 = \frac{\sin 2\alpha_1 \cdot \sin 2\alpha_2 \dots \sin 2\alpha_n}{2^n}$$

$$\text{Maximum of } y^2 = \frac{1}{2^n} ; \text{ maximum of } y^2 = \frac{1}{2^{\frac{n}{2}}}$$

77. A

Sol. Equation of family of plane passing through intersection of X – Y plane and Y – Z plane is $z + \lambda x = 0$ this plane makes an angle 45° with the xy plane ($z = 0$)

$$\cos 45^\circ = \frac{1}{\sqrt{1 + \lambda^2}} ; \sqrt{\lambda^2 + 1} = \sqrt{2} ; \lambda^2 + 1 = 2 ; \lambda = \pm 1$$

Hence, plane are $z + x = 0$; $z - x = 0$

78. A

Sol. Let $\vec{v} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$; $\vec{v} \cdot \vec{a} = 0 \Leftrightarrow x = 0$; $\vec{v} \cdot \vec{b} = 1 \Leftrightarrow y = \frac{1}{|\vec{b}|^2}$

$$\vec{v} \cdot (\vec{a} \times \vec{b}) = z(\vec{a} \times \vec{b})^2 = 1 ; z = \frac{1}{|\vec{a} \times \vec{b}|^2} ; \vec{v} = \frac{\vec{b}}{|\vec{b}|^2} + \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|^2}$$

79. B

Sol. $(p \vee q) \rightarrow (p \vee \sim q)$

p	q	$p \vee q$	$p \vee \sim q$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	T

80. C

Sol. Combined mean $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{80}{5} = 16$

$$d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$$

$$d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$$

$$\text{Combined mean } \sigma^2 = \frac{n_1(\alpha_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2} = \frac{200 \times (9 + 81) + 300(16 + 36)}{500} = 67.2$$

SECTION – B

81. 00403.00

Sol. $\int x(x^{2009} + x^{803} + x^{401})(2x^{1608} + 5x^{402} + 10)^{\frac{1}{402}} dx$
 $\Rightarrow \int (x^{2009} + x^{803} + x^{401})(2x^{2010} + 5x^{804} + 10x^{402})^{\frac{1}{402}} dx$
 Put $2x^{2010} + 5x^{804} + 10x^{402} = t$
 $\therefore a = 403$

82. 00022.00

Sol. $\frac{n}{\sqrt{2}} < 2 \Rightarrow n < 2\sqrt{2} ; n = 1 \text{ or } 2$

Length of chord = $\sqrt{16 - 2n^2}$ the sum = 22

83. 00000.25

Sol. $t^2 = 1 - \frac{2b^2 - a^2}{3b^2 - 2a^2} = \frac{b^2 - a^2}{3b^2 - 2a^2}$; $(f(t))^2 = 1 - \frac{b^2}{2b^2 - a^2} = \frac{b^2 - a^2}{2b^2 - a^2}$

$(f(t))^2 = \frac{t^2}{1 - t^2}$ and $f(t) > 0$; $(f(t)) = \frac{t}{\sqrt{1 - t^2}}$; $f \circ f(t) = \frac{t}{\sqrt{1 - 2t^2}}$

$f \circ f \circ f(t) = \frac{t}{\sqrt{1 - 4t^2}}$; $\int_0^{\frac{1}{2}} f \circ f \circ f(t) dt = \frac{1}{4} = 0.25$

84. 00007.50

Sol. $\int_0^1 (4x^3 f(x) - f^2(x) + 4x^6 - 4x^6) dx = \frac{4}{7}$; $\int_0^a 4x^6 dx - \int_0^1 (f(x) - 2x^3)^2 dx = \frac{4}{7}$

$\int_0^1 (f(x) - 2x^3)^2 dx = 0$ $\therefore f(x) = 2x^3$; $\int_0^2 f(x) dx = \frac{15}{2}$

85. 00002.00

Sol. $\frac{a^2 + b^2 + ab}{a + b} = \frac{a^3 + b^3}{a^2 - b^2}$; $549k = 1098 \therefore k = 2$

86. 00001.00

Sol. $(7 + 10)^{1995} + (1 + 10)^{1995} - (7)^{1995} = (\text{a number of multiple of } 10) + 1$

Unit place digit is 1

87. 00009.00

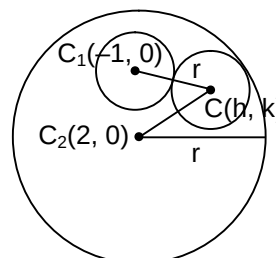
Sol. $A = \begin{bmatrix} 3a & b & c \\ b & 3c & a \\ c & a & 3b \end{bmatrix}$; $\det(A) = 29abc - 3(a^3 + b^3 + c^3)$

$A^T A = 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; $|A^T A| = 4$; $|A|^2 = 4$; $|A| = 2$

$a^3 + b^3 + c^3 = 9$

88. 00013.00

Sol. $CC_1 = r + r_1$
 $CC_2 = r_2 - r$
 $CC_1 + CC_2 = r_1 + r_2$
 Locus of c is an ellipse with focus c_1 and c_2
 $r_1 + r_2 = 2a = 10$
 $|c_1 c_2| = 2ae = 3$
 $e = \frac{3}{10}$
 $p + q = 13$



89. 00007.00

Sol. $f(2) = 0$

$$\frac{dy}{dx} = \frac{(x+1)^2 + y - 3}{x+1}; \text{ let } x+1 = X \text{ and } y-3 = Y$$

$$\frac{dY}{dX} = \frac{X^2 + Y}{X} = X + \frac{Y}{X}; \frac{dY}{dX} - \frac{Y}{X} = X; \frac{d}{dX}\left(\frac{Y}{X}\right) = 1$$

$$\frac{Y}{X} = X + c; (y-3) = (x+1)^2 + c(x+1)$$

$$\text{Put } (2, 0) \quad y = x^2 - 2x$$

$$x = -4 \quad A = \frac{4}{3}$$

90. 00004.00

Sol. $N_1 = 10^{83} = 2^{83} 5^{83}$

$$N_2 = 10^{70} = 2^{70} 5^{70}$$

So, divisor are of type $2^a 5^b$

$$0 \leq a, b \leq 83 \text{ (for } N_1)$$

$$\text{For multiple of } 10^{70} \Rightarrow 70 \leq a, b \leq 83$$

$$\text{Probability} = \frac{14 \times 14}{84 \times 84} = \frac{1}{36} \therefore \alpha = 36$$